

TASK 1 – ANALYSE

Scrum Element	Still Works?	Why / Why Not?
2-week sprints	Yes, mostly	Development work can still be organized into short iterations. However, ML training, data collection, clinical validation, and regulatory reviews may take longer than one sprint.
Definition of Done	Partly	Traditional software criteria (tested, documented, deployed) are insufficient. The DoD must include model performance metrics, safety validation, documentation, and compliance requirements.
Backlog prioritization	Yes	Prioritization remains useful, but regulatory, privacy, and safety tasks become as important as technical features.
Sprint Review Demo	Partly	Normal feature demonstrations work, but ML outputs are probabilistic and may require statistical evaluation rather than simple demonstrations.
Velocity tracking	Less useful	Story points measure development effort but not model quality, clinical safety, or regulatory progress. A team may have high velocity while still being far from approval.

New Problems Introduced by ML

- Model behavior depends on training data rather than code.
- Outputs are probabilistic and can be wrong.
- Incorrect dosage recommendations can cause severe patient harm.
- GDPR compliance is required for sensitive health data.
- MDR approval introduces significant documentation and validation work.
- Model performance may degrade over time and requires monitoring.
- Team lacks ML expertise.
- Different patients require individualized recommendations.

TASK 2 – DECIDE

Situation A – Who Do You Hire First?

a. Which 2 do you hire first?

1. ML Engineer
2. Regulatory Specialist

Justification

The ML Engineer is essential because the company currently has no capability to build or train the model.

The Regulatory Specialist is critical because MDR requirements affect architecture, documentation, testing strategy, risk management, and clinical evidence from the beginning. Regulatory mistakes discovered later could require extensive rework.

b. What risk do you consciously accept?

By delaying:

Clinical Validator

- Risk of developing a model that later proves medically unsafe.

Data Privacy Officer

- Risk of discovering GDPR compliance issues after data collection plans are already established.

c. Earliest bottleneck?

- Data Privacy Officer becomes critical before any patient data is collected or processed.
- Clinical Validator becomes critical once a working prototype exists and medical validation begins.

Situation B – Which Data Do You Use?

a. Which dataset?

Hospital B (5,000 records with informed consent).

Justification

Patient consent and GDPR compliance are non-negotiable. Using Hospital A's data could create legal and

ethical problems that may prevent approval entirely.

b. Consequences for patient safety

Positive:

- Data is legally and ethically obtained.
- Greater trust from regulators and patients.

Negative:

- Smaller dataset may reduce model accuracy.
- Increased risk of missing rare edge cases.

c. Consequences for project timeline

- Additional time may be needed to collect more consented data.
- Model development could progress more slowly.
- Regulatory approval is more likely because the data source is legally defensible.

Situation C – What Do You Tell the CEO?

a. Which option?

Option 2: Build a Shadow Mode Prototype

Justification

This approach respects MDR regulations because patients never see the recommendations.

At the same time, doctors can evaluate the model's outputs and generate evidence for future approval. The company can still demonstrate technical progress to investors without committing a criminal offence.

b. Project Timeline

MonthActivity

0–3 Hire specialists, establish data governance, acquire consented data
3–6 Build first ML prototype and shadow-mode system
6 Demonstrate shadow-mode prototype to investors and clinicians
6–12 Clinical validation and safety testing
12–18Regulatory documentation and MDR approval process
18+ Approval obtained and patient-facing release begins

TASK 3 – REFLECT

a. Hardest situation?

Situation A was the hardest because all four roles are important and delaying any of them creates significant risk. Choosing only two requires balancing immediate technical progress against future compliance and safety concerns.

b. What would change your answer?

If the existing team already included experienced regulatory experts, I would hire the Clinical Validator instead of the Regulatory Specialist. In that case, patient safety validation would become the larger capability gap.

AI ANALYSIS

a. Did the AI make a valid counterargument?

One strong counterargument is that hiring the Data Privacy Officer before the Regulatory Specialist could prevent GDPR mistakes that may contaminate the entire dataset. If patient data is collected incorrectly, even a technically successful model may become unusable.

b. Would you change any choices?

I would still keep the Shadow Mode decision for Situation C because it best balances innovation with legal compliance. However, after considering the counterargument, I might reconsider Situation A depending on how much external regulatory consulting is available during the early project stages.

